

All 55 Grokking runs: τ_{gen} VS τ_F

G<F: 52/55 | F<G: 2/55 | coincident: 1/55

$F < G$ ($\tau_F < \tau_{\text{gen}}$)

Fourier geometry onset τ_F ($\times 10^3$ steps)

- Stage 2 (wd sweep)
- Stage 3 (lr sweep)
- $F < G$ runs (★)
- $\tau_F = \tau_{\text{gen}}$

Generalization onset τ_{gen} ($\times 10^3$ steps)

